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Promoting knowledge recommendation in innovative engineering design: a BERT-GAT-based patent representation learning approach

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ABSTRACT

Since innovation in complex product design hinges on thorough engineering knowledge application, high-quality patent recommendations foster innovation in engineering design. However, many patent knowledge recommendation studies perform patent analysis without comprehensive exploration and proper organisation of knowledge, causing a superficial understanding of patents and returning arbitrary results. To mitigate this issue, a deep learning-based approach for patent representation learning and knowledge recommendation is proposed. First, a four-dimensional patent knowledge model is defined to formalise the patent attributes that critically affect the engineering design outcomes, namely patents' domain(D), function(F), technology(T) and citation(C). Second, to exploit patent knowledge from their content and citation relationships, a representation learning approach integrating Bidirectional Encoder Representations from Transformers(BERT) and Graph Attention Network(GAT) is introduced. Thereafter a patent knowledge space is established in which each patent is characterised by the function, technology, and citation embeddings. Third, a knowledge requirement space is also constructed by vectorising a designer's search query via BERT model and linking it to a requirement-representing patent based on similarity. Finally, a recommender prototype is developed and showcased by the knowledge recommendation in sealing structure design tasks. Comparative experiments and application cases validate the effectiveness of our method in patent representation learning and knowledge recommendation.

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Innovative engineering design; patent representation learning; knowledge recommendation; BERT; GAT

1. Introduction

The engineering design of complex engineering products featured with sophisticated structures, such as self-driving vehicles, airplanes and manned spacecrafts, is a knowledge-intensive and technology-driven process, requiring a substantial support of domain knowledge (Li, Yi, and Cao 2022b). Meanwhile, because the complex products are low volume and customised, specific requirements need to be met in the design stage (Zhang et al. 2023).

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However, while facing the knowledge-intensive and requirement-demanding design task, designers' limited knowledge is often insufficient to support the entire design task, resulting in a low level of innovativeness and a long problem-solving process (Wang et al. 2022). Regarding such a problem, automatically retrieving knowledge and recommending it to designers based on diverse design requirements would exert a catalysing effect on innovative engineering design.

Patents, a wealth of technologies across almost all domains, could inspire innovative design ideas by deploying the ingenuity embedded within them (Soo et al. 2006). Since the number of granted patents has grown to a huge figure in recent years (Trappey, Trappey, and Hsieh 2021), existing studies typically transform patents directly into computable representations for large-scale investigation. Nevertheless, referable knowledge within patents is multifaceted and highly dispersed, often mingled with irrelevant information. Directly analyzing patents for recommendation without a proper organisation of these knowledge would suffer from the defect of providing perplexing results that difficult to be interpreted. To mitigate this issue, establishing a knowledge model for different types of patent knowledge to enable ordered organisation and targeted analysis would contribute to obtaining more reasonable recommendation results.

Currently, the most frequently used patent knowledge mining methods, such as topic modelling (Govindarajan, Trappey, and Trappey 2019), dependency parsing (Jang, Jeong, and Yoon 2021) and named entity recognition (NER, Chen et al. 2020), mainly focus on the semantic analysis of patent texts. However, simply extracting knowledge from patent text may overlook a large volume of valuable information, leading to an insufficient mining of patent knowledge. Meanwhile, simply depending on patent texts for recommendation may lead to a surplus of highly similar patents in the results, causing redundancy and hindering the generation of sufficiently innovative design ideas. The citation information of patents represents reference relationships among technologies, indicating that one technology has contributed to the development of another technology, thus containing certain knowledge as well (Érdi et al. 2013). Within patent citation relationships, knowledge is no longer limited to a single technology but exists in a macroscopic form across various domains, serving as a complement to the knowledge in patent texts. By incorporating patent citation relationships into the consideration of patent knowledge recommendation, patents that are significantly valuable but highly different in text can be discovered, and this offers designers recommendation results with more diversity. To the authors' best knowledge, few studies fully exploit the patent citations for knowledge recommendation, with the majority treating citation information merely as evaluation metrics (Li et al. 2012; Qiu and Wang 2023). Moreover, integrating knowledge from patent texts and citation relationships is crucial for achieving formalised representations of patent knowledge. Therefore, it is necessary to develop an approach that comprehensively explore and integrate knowledge from both patent texts and citation relationships to enhance the effectiveness of knowledge recommendation.

As a powerful representation learning technique, deep learning has significant advantage in processing massive and complex information, and it has already found applications in patent analysis studies (Trappey, Liang, and Lin 2022; Qiu and Wang 2023). Applying deep learning techniques for patent knowledge representation can effectively capture the latent features of patents as well as the network formed by citation relationships, thereby discovering more valuable information.

In this study, we propose a deep learning-based patent representation learning approach that integrates Bidirectional Encoder Representations from Transformers (BERT, Devlin et al. 2019) and Graph Attention Network (GAT, Veličković et al. 2018) along with normalisation techniques to derive comprehensive patent knowledge representations from patent texts and citation relationships for knowledge recommendation. BERT is a deep learning-based pre-trained language model that exhibits significant advantage in complex semantic comprehension and word representation, and GAT is an attention mechanism-included graph neural network which can learn the importance of different citation relationships while integrating the citation network. The goal of the proposed approach is to provide designers with suitable patent knowledge that meets their requirements and facilitate the generation of innovative design ideas. The contributions of this study can be outlined in the following points.

- A four-dimensional patent knowledge model for a formalised organisation of different types of patent knowledge.
- A BERT-GAT-based patent knowledge representation learning approach for comprehensively exploiting and integrating knowledge from both patent texts and the citation network.
- A recommender system prototype with two recommendation schemes for effectively stimulate innovative ideas of designers.

The rest of the article is organised as follows. The literature of knowledge recommendation in engineering design and patent representation learning approaches for this scenario as well as two related techniques are summarised in Section 2. In Section 3, a four-dimensional patent knowledge model is first defined, followed by the establishment of a patent knowledge space and a knowledge requirement space, and the BERT-GAT-based patent representation learning approach is expounded. Two knowledge recommendation schemes that one based on patent similarity solely and the other on a cross-domain indicator are also presented in this section. Two experiments for verifying the effectiveness of the proposed patent representation learning approach are performed in Section 4. Section 5 demonstrates two application cases about the innovative design of sealing structures to illustrate the feasibility of the representation learning approach through the performance of knowledge recommendations. In Section 6, the contributions and future research directions of this study are concluded.

2. Related works

2.1. Knowledge recommendation in engineering design

The purpose of knowledge recommendation in engineering design is to trigger the creativity of designers and thus generate innovative ideas (Yang, Quan, and Zeng 2022). Predominantly adopted approaches of knowledge recommendation encompass user preference-based approaches, expert systems, knowledge graphs (KGs) and representation learning techniques.

User preference-based approaches that mines users' preferences from reviews or comments to products can realise personalised knowledge recommendation. Wang et.al. (2018)

established affective profiles for products using online product descriptions and user reviews to visualise the gap between manufacturers' intentions and users' responses, thereby assisting in product developments and improvements. Despite its significance in the field of normal commercial products, such approaches still encounter limitations when applied to complex engineering product design situations, primarily due to their reliance on massive user feedback.

Expert systems possess the capacity to simulate cognitive processes and decision-making patterns of domain experts, offering reasoned recommendations. Drawing upon a knowledge base composed of rules, object-oriented constructs, component technical data, and procedural functions, Matelli, Bazzo, and Da Silva (2009) developed an expert system prototype to assist the conceptual and preliminary design of natural gas cogeneration plants. Kim, Roh, and Ha (2015) built an expert system for the arrangement design of a submarine, and design alternatives could be evaluated according to past data, expert experiences and design rules. Yang, Dai, and Jiang (2023) established an expert system tailored for additive manufacturing design solutions through the combination of assembly structure characteristics and decision-making algorithms. Due to the ability to store and reason over domain knowledge, expert systems can offer highly specialised assistance in engineering design scenarios. However, the construction of an expert system requires a substantial investment of time and manual labour for knowledge acquisition and management, restricting its application to a relatively narrow scope.

Knowledge graphs are structural knowledge networks with rich semantics, which can enhance the in-depth understanding and facilitate knowledge recommendation by modelling various knowledge elements as well as their attributes and relationships in a graphical form. Hao et al. (2021) realised rapid and accurate decision-related knowledge navigation based on a meta-model of decision knowledge graph, which is grounded in the compromise Decision Support Problem. Jia, Zhang, and Saad (2022) represented the observable design result facts of products using design knowledge graphs to effectively capture and reuse tacit design knowledge for new solutions, and the target solution could be determined through relational learning. Li, Wang, and Rong (2023) constructed a product fault knowledge graph based on product operation and maintenance data, which can provide guidance for the design of next-generation products through relationship mining between nodes and risk calculation on different parts. The application of knowledge graphs for knowledge recommendation has demonstrated excellent knowledge searching efficiency and the capability of discovering implicit knowledge, significantly advancing the integration and reuse of technical knowledge. Nevertheless, in the highly specialised field of complex engineering product design, constructing high-quality domain knowledge graphs is still facing certain challenges, such as knowledge modelling and data quality issues.

Representation learning refers to a set of techniques that can automatically capture the inherent features of data to obtain more informative representations. In the process of design knowledge recommendation, applying representation learning techniques can better understand the features of problems and knowledge, enabling more precise recommendations (Lee and Sohn 2021; Trappey, Liang, and Lin 2022). Unlike alternative methodologies, representation learning approaches do not rely on static knowledge

bases, thus demonstrating better adaptability and generalisation ability to new knowledge, and consequently more suitable for knowledge recommendation in engineering design.

2.2. Patent representation learning for knowledge recommendation

Patents contain rich knowledge that can assist in engineering design, and introducing representation learning approaches into patent analysis facilitates the comprehensive extraction of technical features, ultimately improving the performance of knowledge recommendation. Liu, Hsieh, and Huang (2022) represented patents as keyword vectors generated by Doc2vec algorithm and calculated the similarities with evolutionary benefits related to TRIZ theory, thereby screening relevant patents applicable for efficient problem solving in product design. To provide a recommendation system for technology convergence opportunities, Lee and Sohn (2021) trained a self-supervised neural network to learn latent representations of convergence items and firms, which are implicitly embedded within patent's co-classification information. Similarly, another patent recommendation system for technology trend mining was developed by Trappey, Trappey, and Hsieh (2021), in which the textual contents of patents were represented as vectors via a neural network language model. Deep learning models (e.g. BERT) have superior performance in patent representation learning, enabling the extraction of more complex information for knowledge representation and recommendation (Trappey, Liang, and Lin 2022; Qiu and Wang 2023).

Among these representation learning-based approaches, patent text or other information is usually processed directly by the model. Nevertheless, patent knowledge is of various types and forms, and often mixed with irrelevant information. Directly conducting representation learning on patents often faces problems in covering various forms of patent knowledge, and distinguishing between different types of knowledge and irrelevant information is also challenging. This complexity makes it difficult to achieve a clear match with designers' knowledge requirements during the recommendation process. For this reason, this study divided patent knowledge into different dimensions for a clear organisation in patent representations, and employed deep learning-based term extraction and graph representation learning techniques to separate different dimensions of patent knowledge from irrelevant information and exploit the knowledge source in graphical form (i.e. the citation network), respectively.

2.3. Term extraction techniques in patent mining

In existing patent analysis research, widely adopted methods for term extraction include term frequency-based approaches, syntactic analysis and NER. Compared to simply calculating word frequency, term frequency-inverse document frequency (TF-IDF) can evaluate the importance of terms within patents more comprehensively. Due to the lack of semantic representation capability, this approach is usually used as a supplementary measure in combination with other approaches for various tasks such as semantic similarity calculation (Shen, Wang, and Yang 2020), weight assignment (Kim et al. 2019) and performance evaluation (Kim and Yoon 2022).

Syntactic analysis approaches have an excellent ability in extracting patent knowledge from particular types, since different types of knowledge in patent texts typically exhibit distinct semantic features. To facilitate cross-domain knowledge recommendation, Liu et al. (2020) established a patent knowledge space with function, technology and domain properties of patent knowledge, and corresponding technical terms were extracted by semantic evaluation indices. Yu et al. (2022) utilised Subject-Action-Object (SAO) structure to extract technical information from patent texts and introduced multiple technical indicators to screen cross-domain patents, thereby assisting in radical innovative design. However, patent texts are characterised by complicated sentence structures, dense technical terms and language ambiguity, while semantic analysis usually focuses merely on particular syntactic structures determined by rules and indices. Therefore, such approaches still have limitations in the accuracy and comprehensiveness of patent knowledge mining.

NER is a method for identifying rigid designators belonging to distinct categories from text and performs well in extracting domain-specific feature terms (Li et al. 2022a). In patent analysis, these rigid designators could be technical terms from different types of patent knowledge. In current studies, NER is mainly conducted using rule-based approaches (Giordano et al. 2023) or distributed representation learning-based approaches (Chen et al. 2020). Since rule-based approaches requires the manual definition of rules, the handling of highly specialised and complex patent knowledge is greatly impacted by the level of expertise of the individuals, and the efficiency is also limited. In comparison, distributed representation learning-based approaches automatically learn feature representations from massive data, which avoids the laborious process of manually defining rules. Among various distributed representation learning-based NER approaches, BERT exhibits superior performance in word representation and context comprehension, and has been validated to be applicable in various NER-included engineering scenarios. Li et al. (2021) employed BERT-based NER to extract flat and nested entities from bridge inspection texts. Yu et al. (2022) built a mineral named entity recognition model based on BERT to support the construction of mineral knowledge graphs. Despite Puccetti et al. (2023) performing BERT-based NER on patent texts for technology identification, this approach has not yet been widely applied in patent mining. For the above reasons, we adopted BERT-based NER technique to extract and represent patent knowledge originate from textual contents.

2.4. Graph-based approaches for patent representation learning

To gather more information from patent documents, graph-based approaches have been proved to have outstanding performance. Zhai et al. (2022) constructed a patent heterogeneous network using entities of function, solution and application field extracted from patent texts, and developed a graph embedding algorithm for the network to exploit detailed and extensive patent information. In addition to textual information, graph-based approaches also enable in-depth exploration of various other valuable information. Sidharth, Li, and Luo (2022) apply TransE algorithm to patents' citation network and inventor network respectively, and these graph embeddings are further combined with patent text embeddings to enhance patent-to-patent retrieval. To enhance automatic patent classification, Fang et al. (2021) proposed a novel patent representation learning approach named Patent2Vec according to a multi-view analysis of patent text and metadata. Jiang and Luo

(2022) generated deep neural embeddings of patents' citation network by applying GraphSAGE algorithm, and a technology fitness landscape for guiding technological innovation was established.

In the studies mentioned above, apart from the newly proposed algorithms, commonly used models such as TransE and GraphSAGE have a common limitation in the representation learning for citation networks, specifically, the inability to consider the importance of different citation relationships. Due to differences in the motivations behind citation relationships between different patents, some citation relationships contain richer and more meaningful knowledge than others. Therefore, taking the importance of different citation relationships into consideration will have a positive impact on a more comprehensive patent representation learning. In this regard, GAT is a preferable option, since its attention mechanisms effectively consider the importance of different edges between nodes in graph representation learning. As a result, beyond BERT-based NER, this study further applies the GAT model for representation learning of the patent citation network.

3. Methodology

The overall workflow of the proposed methodology for patent knowledge representation and recommendation is presented in Figure 1. With a pre-defined knowledge model, formalised representations of the patents are derived through a BERT-GAT-based representation learning process of both the textual contents and citation network, thereby forming a patent knowledge space. In addition, a knowledge requirement space is established by matching designers' query to a requirement-representing patent interactively. Finally, effective knowledge recommendation is realised by mapping the knowledge requirement space to the patent knowledge space based on similarity, and a cross-domain knowledge recommendation scheme is also proposed.

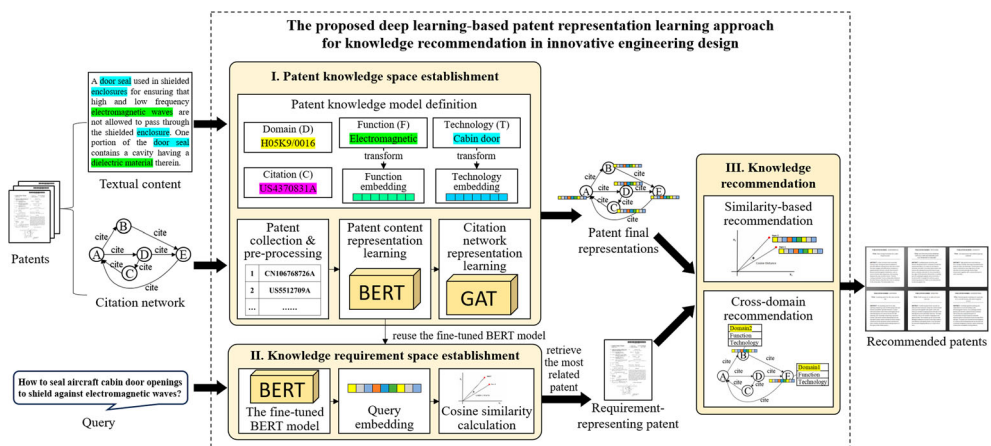


Figure 1. The overall workflow of the methodology.

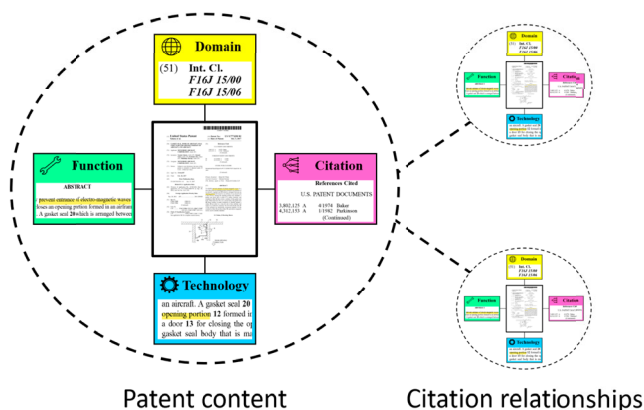


Figure 2. Illustration of the four-dimensional patent knowledge model.

3.1. Patent knowledge space establishment

Knowledge contained in patents is heterogeneous and multi-dimensional, being depicted by explicit and implicit information from various aspects. As the knowledge is not originally presented in a structured form, hindering direct analysis and recommendation, a four-dimensional patent knowledge model is defined to represent the key features of patent knowledge, as shown in Figure 2. Since any patent could be supported by such a knowledge model, a collection of patents could form a four-dimensional patent knowledge space.

According to Liu et al. (2020), the content of a patent contain function, technology and domain properties. In this paper, we adopted International Patent Classification (IPC) system to indicate the Domain dimension of the patent knowledge model, since it is a widely accepted hierarchical classification framework for patents' technical fields developed by World Intellectual Property Organisation (WIPO). For the Function dimension, we refer to Choi et al.'s (2013) definition and describe it as the purposes and effects that the patented technology can achieve, which are presented as terms. In the textual contents of different patents, information representing technical features generally varies in granularity and expression, leading to difficulties in directly determine what the Technology dimension refers to. Considering that similar technical features could be deployed in similar situations, we substitute the application situation of the patented technology for technical features themselves, serving as the Technology dimension of the patent knowledge model, and also presented as technical terms.

Because patent citation relationships contain valuable technical knowledge that can hardly be reflected by textual contents, in addition to the above three dimensions, our patent knowledge model has also incorporated a Citation dimension.

3.1.1. Patent dataset preparation

The patent dataset consisting of textual contents and citation relationships could be built using online patent search platforms. To ensure the inclusion of sufficient citation relationships that contribute to a larger coverage of technical domains, we carry out a three-hop-citing patent retrieval scheme: 'Hop 1 patent – [cite] – Hop 2 patent – [cite] – Hop 3 patent',

specifically, collecting an initial set of patents (Hop 1 patents) with their citation lists manually, and gathering the subsequent hops of patents according to the citation lists of the preceding hops of patents. The content of each collected patent includes publication number, title, abstract and IPC codes. Based on the citation relationships, the collected patents can form a citation network, where each node represents a patent and each edge represents a citation relationship.

However, raw patent texts are typically characterised by language inconsistency and numerous special marks, and some earlier patents have the problem of missing contents. Therefore, with a view to enhance the format uniformity and semantic clarity of patent texts in the dataset, an appropriate data pre-processing step is necessary, and the procedure will be elaborated in detail in Section 4.1.

3.1.2. Representation learning of patent contents

With the aim of obtaining patent representations that fully reflect the pre-defined knowledge dimensions in patent texts, a BERT model is employed with NER and weighted averaging techniques, thereby generating feature vectors and tags for the patents, as illustrated in Figure 3. Considering that the functions and technical characteristics of patents are presented as terms, while domain information can be indicated through IPC codes, this process is applied only to the representation of the Function dimension and Technology dimension, and the patent contents used in this process are patent titles and corresponding abstracts, which are concatenated.

As a deep learning-based language model pre-trained by large-scale textual datasets, BERT has learnt rich semantic and contextual information of natural language, and can be fine-tuned for various downstream tasks, such as question answering, sentiment analysis and NER. The tokenisation of the pre-trained BERT model is performed by Wordpiece tokeniser (Wu et al. 2016), which can turn sentences into wordpieces, and an example is given in Figure 4. Compared with sentence – or word-level embedding methods, Wordpiece tokeniser enables the BERT model to generate embeddings at a finer granularity, enhancing its ability in capturing the internal structure, semantic relationships and contextual associations of words. In this paper, after being fine-tuned on an NER task, a BERT model suited for patent text representation and term extraction could be derived, and the pre-processed patent texts could be turned into wordpiece-level token embeddings by this model.

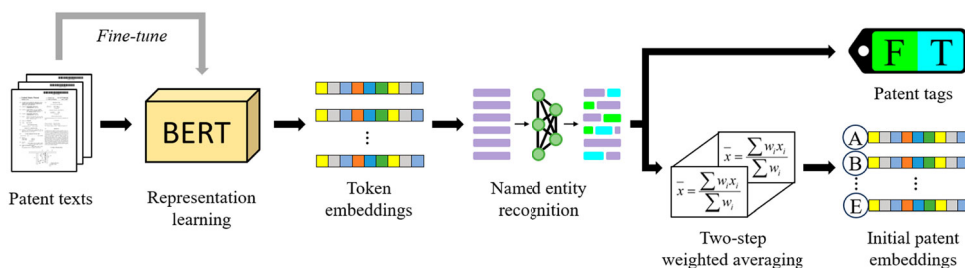


Figure 3. BERT-based representation learning process of patent contents.

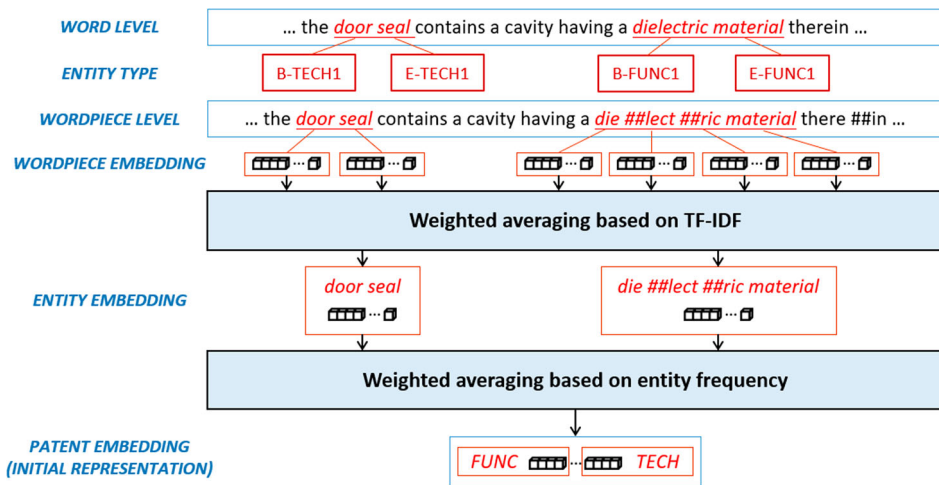


Figure 4. An illustration of Wordpiece tokenisation and the two-step weighted averaging approach.

To extract functional or technical terms as features of the patents, NER is subsequently performed on the token embeddings. The NER process is essentially a classification process for token embeddings. When a token embedding is identified as belonging to a Function or Technology entity, the position of the corresponding token in the sequence could be determined, and the entity term could be extracted. Different from conventional term extraction approaches based on word frequency, terms extracted from the NER process can semantically represent the features of patents since token embeddings from the fine-tuned BERT model have comprehensively captured contextual information. After the NER process, the most frequent Function and Technology entity categories in each patent are assigned as the patent's feature tag, expressed as 'Function-Technology'.

Another utilisation of the token embeddings is for patent text representation. Due to the varying lengths of patent texts, different patents may correspond to different numbers of token embeddings. For the purpose of obtaining formalised patent representations, a two-step weighted averaging approach is introduced to convert the token embeddings of entities in each patent into a single patent vector, as shown in Figure 4.

Since the tokens of Function and Technology entities are determined in the foregoing NER process, in the first step, a weighted averaging process based on TF-IDF is implemented to merge the embeddings of these entity tokens into entity-level embeddings. For each entity, its embedding is calculated by Equation (1).

$$entity_embedding_j = \frac{\sum_{i=1}^m token_embedding_i \times TFIDF_i}{\sum_{i=1}^m TFIDF_i} \quad (1)$$

where i refers to each token, m is the number of tokens within the entity, and the TF-IDF based weight for each token embedding is calculated by Equation (2).

$$TFIDF_i = \frac{N_i}{N_w} \times \log \frac{N_p}{N_p(i)} \quad (2)$$

where N_i is the frequency the token appears in the patent, N_w is the total number of word-piece tokens in the patent, N_p is the number of patents in the dataset and $N_p(i)$ is the number of patents containing the token. By applying TF-IDF values as weights, the impact of meaningless wordpieces within an entity is reduced, and the functional and technical features are thus strengthened.

With the token embeddings merged into entity-level embeddings, in the second step, these embeddings are further merged into patent-level embeddings through an entity frequency-based weighted averaging process. Specifically, the normalised embeddings of Function entities and Technology entities are calculated separately, and then concatenated as the patent embedding. This process is expressed by Equations (3–5).

$$F_embedding = \frac{\sum_{j_f=1}^{n_f} entity_embedding_{j_f} \times Freq_{j_f}}{\sum_{j_f=1}^{n_f} Freq_{j_f}} \quad (3)$$

$$T_embedding = \frac{\sum_{j_t=1}^{n_t} entity_embedding_{j_t} \times Freq_{j_t}}{\sum_{j_t=1}^{n_t} Freq_{j_t}} \quad (4)$$

$$patent_embedding = F_embedding || T_embedding \quad (5)$$

where j_f and j_t respectively refer to each Function and Technology entity, n_f and n_t is the number of the two types of entities within the patent, $Freq_{j_f}$ and $Freq_{j_t}$ is the frequency of each entity appearing in the patent, and $||$ represents concatenation.

However, not all of the patents simultaneously contain Function and Technology entities. For the missing entity types, we assign their embeddings ($F_embedding$ or $T_embedding$) as zero vectors. Thus far, every patent is matched with a fixed-length vector, with formalised representations of functional and technical features.

3.1.3. Representation learning of the citation network

Regarding the representation of the Citation dimension of the patent knowledge model, given that all citation relationships form a citation network, we introduced a GAT model to integrate citation information into patent representations, which is shown in Figure 5. GAT is a powerful deep learning architecture for analyzing graph-structured data. The inclusion of attention mechanisms allows the GAT model to weigh the importance of different neighbours for each node individually, enabling it to capture complex relationships within the graph. In this study, GAT is utilised to aggregate information from citation-neighbouring patents to each target patent representation based on learned weights, thereby integrating patent knowledge in the citation network.

During the preceding patent text representation process, different numbers of Function and Technology entities are identified from each patent. For patents that containing both Function and Technology entities, we create a sub-dataset using their embedding vectors, feature tags and citation relationships among them, which is utilised for training the GAT model through a node classification task. With the well-trained GAT model, the feature vector of each patent could be updated based on the features of its citation neighbours. After being mapped to a high-dimensional space by the learned weight matrix W , the attention coefficient e_{ij} between each target patent node i and a citation-neighbouring node j could be calculated by Equation (6; Velićković et al. 2018), representing the importance of this

citation relationship to the target patent node i .

$$e_{ij} = \text{LeakyReLU}(\vec{a}^T [W\vec{v}_i || W\vec{v}_j]) \quad (6)$$

where $||$ refers to the concatenation of two vectors, $\vec{v}_i$ and $\vec{v}_j$ are the feature vectors of patent node i and j respectively, $\vec{a}$ is a learned weight vector. Subsequently, the attention coefficient e_{ij} is normalised by the softmax function, transforming into a standardised attention weight α_{ij} , as shown in Equation (7; Veličković et al. 2018).

$$\alpha_{ij} = \text{softmax}(e_{ij}) = \frac{\exp(e_{ij})}{\sum_{k \in \mathcal{N}_i} \exp(e_{ik})} \quad (7)$$

where $\mathcal{N}_i$ refers to patent node i 's citation neighbourhood. After this step, the standardised attention weights are utilised for a weighted summation of the features of all citation-neighbouring nodes. Furthermore, a multi-head attention mechanism is adopted to more comprehensively capture the intricate relationships among patent nodes by applying multiple groups of independent attention mechanisms, and the updated feature vector $\vec{v}'_i$ of the target patent node i is yielded. This process could be expressed as Equation (8; Veličković et al. 2018).

$$\vec{v}'_i = \sigma \left(\frac{1}{H} \sum_{h=1}^H \sum_{j \in \mathcal{N}_i} \alpha_{ij}^h W^h \vec{v}_j \right) \quad (8)$$

where h refers to each independent attention mechanism, H is the number of attention mechanisms, σ is an activation function and $\vec{v}'_i$ is the updated feature vector of the target patent node i . Up to this point, citation information is incorporated into the patents' feature vectors, and the final representations of the patents are derived.

3.2. Knowledge requirement space establishment

In general, when faced with a design problem, designers will generate certain knowledge requirements. While solving the design problem, these knowledge requirements are often

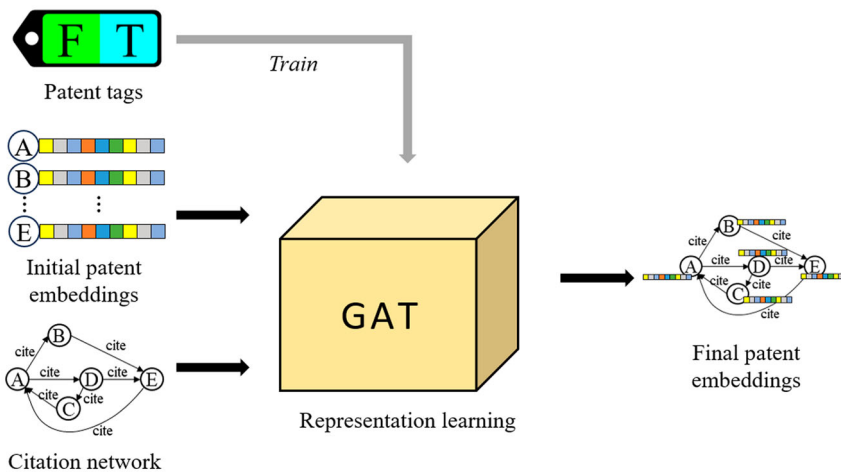


Figure 5. GAT-based representation learning process of the citation network.

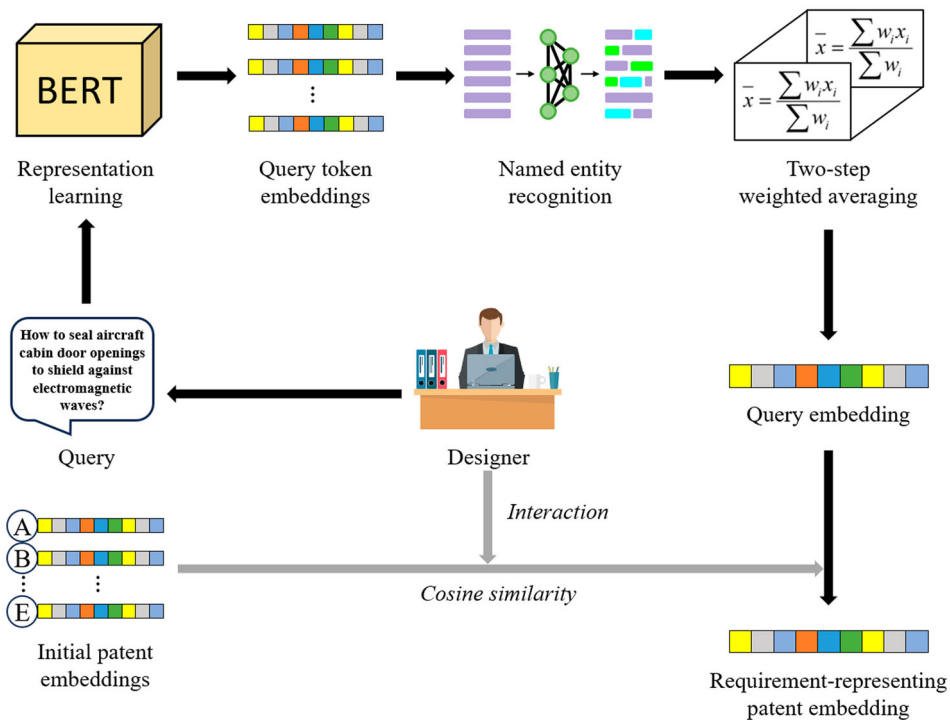


Figure 6. Process of establishing the knowledge requirement space.

expressed as a search query in natural language for searching, which also include features of three knowledge dimensions, i.e. Domain, Function and Technology. Herein, a knowledge requirement space is established to facilitate an effective matching of the knowledge requirements with patent knowledge, thereby recommending more enlightening results.

The establishment of the knowledge requirement space is similar to the representation learning process of patent textual contents in their main procedures, which is illustrated in Figure 6. After encoded by the BERT model, the query is converted into a sequence of token embeddings. Then, NER is performed on these vectorised tokens, and the two-step weighted averaging process is implemented to merge the entity token vectors into a single query vector, which is the same as the representation of a single patent. Nevertheless, the query vector involves only the Function dimension and the Technology dimension, and cannot be linked to the higher-dimensional patent knowledge space directly. Considering that all the patents have representations in both two dimensions (Function and Technology) and three dimensions (Function, Technology and Citation), an interactive approach is developed, motivated by the idea of representing knowledge requirements by a patent.

By applying Equation (9), the cosine similarity between the query vector and each initial patent embedding vector is calculated respectively and ranked in descending order. Following this order, the patents with the highest rankings are pushed to designers, and the designers can pick out the most relevant patent to their knowledge requirements, designating it as the requirement-representing patent. This interactive step mitigates the

problem of low performance in knowledge recommendation resulting from imprecise expressions by designers, since the designers have the ability to decide whether the recommended knowledge is indeed relevant to their knowledge requirements. Through this approach, the knowledge requirements of designers are represented in the same dimensions as the patents' final representations, and the knowledge requirement space is thus established.

$$\text{cosine_similarity} = \frac{(\text{vec}_1, \text{vec}_2)}{\text{vec}_1 \text{vec}_2} \quad (9)$$

3.3. Mechanisms for knowledge recommendation

3.3.1. Similarity-based knowledge recommendation

The essence of knowledge recommendation lies in transferring knowledge requirements to a knowledge repository through a mapping process, ultimately selecting the most appropriate results for output. Given that the knowledge requirement space and the patent knowledge space are formed by the requirement-representing patent and patents' final representation vectors respectively, the mapping process could be established to achieve knowledge transfer. As the requirement-representing patent serves as a formalised representation of the knowledge requirements across Function, Technology and Citation dimensions, which is consistent with the patent knowledge space, its cosine similarity with every other patent could be calculated and ranked. In the knowledge recommendation process, patents with the highest similarity values are pushed to designers as the most relevant matches to the requirements in terms of the three knowledge dimensions.

3.3.2. Cross-domain knowledge recommendation

During the design process of some specific engineering products, existing technical documents may suffer from issues of limited quantity and solution diversity, which poses difficulties in providing designers with sufficient assistance. In such situations, recommending knowledge with similar technical features from other domains often has the potential to stimulate designers' creativity, leading to a significant improvement in the innovativeness of the design.

In the patent knowledge model, the Domain dimension is indicated by IPC codes, and each patent's IPC codes could be expressed as a list. Between two patents, a lower similarity of their IPC code lists means a greater difference in their domains. To locate the results for cross-domain knowledge recommendation, a cross-domain patent judging step using patents' IPC lists is added to the similarity-based recommendation process. Following the descending order of cosine similarity, the Jaccard similarity of the requirement-representing patent's IPC code list and that of each other patent are calculated according to Equation (10).

$$J(I_A, I_B) = \frac{I_A \cap I_B}{I_A \cup I_B} \quad (10)$$

where I_A and I_B refer to the IPC code lists of two patents. By applying a pre-assigned threshold, the patents with higher cosine similarity and a Jaccard similarity lower than the threshold are selected for cross-domain knowledge recommendation.

4. Experiments

This section describes the preparation process of the proposed patent representation learning approach and conducted two experiments for validation. Firstly, a patent dataset with the topic of sealing technology was created based on the above-mentioned three-hop-citing retrieval scheme. Then we introduced the training process of the models in the proposed approach and baseline approaches. To verify the performance of the proposed approach, two experiments including patent classification and citation prediction were conducted. The patent classification experiment aims to assess the representation ability for Function and Technology features in patent contents, and the citation prediction experiment aims to assess the representation ability for the citation network.

4.1. Patent collection & pre-processing

With the three-hop-citing patent retrieval scheme, we first gathered 100 sealing-related patents with their citation lists from Google Patents manually, assigning them as 'Hop 1' patents. Based on these citation lists, we sequentially retrieved 1414 'Hop 2' patents and 16,139 'Hop 3' patents from various public patent retrieval platforms, including Google Patents, Espacenet, Patentscope and CNIPA Patent Search and Analysis System, and this process was supported by a data crawler named Octoparse. To ensure the closure of the citation network, a cleanup of the citation lists of 'Hop 3' patents was performed so that patents that not in our dataset were removed.

The data pre-processing step consists of incomplete patent removal, translation and special mark cleanup. For patents with missing titles, abstracts or IPC codes, we directly removed them from the dataset, along with the citation relationships linked to them, thereby reducing the dataset to 15,407 patents with 55,925 citation relationships. To address the language inconsistency problem arising from various nationalities of the patents, we translated all the titles and abstracts into English through Google Translate. In addition, patent texts often contain numerous special marks, such as '<IMAGE>', 'COPYRIGHT' and reference numbers for the figures. Since these marks have a negative impact on the semantic context, we employed a set of regular expressions to remove them. Lastly, for the convenience of tokenisation, we separated all the punctuations from words by spaces.

4.2. Model preparation

4.2.1. BERT

The training dataset used for fine-tuning the BERT on an NER task comprises titles and abstracts from 400 patents, with each word associated with a tag for classification. The tags were assigned by experts, which include 7 Function entities and 8 Technology entities. When tagging each word, we applied the BIOES scheme (e.g. 'S-Function_Leakproof'), where B indicates the beginning of an entity, I indicates that the word is inside an entity, O indicates that the word does not belong to any entity, E indicates the end of an entity, and S indicates that the word is a single-word entity. This training dataset was saved in a TSV file after tagging, with each row consists of a word in the first column and its corresponding

tag in the second column. During the tokenisation by Wordpiece tokeniser, the tag of each token was consistent with its source word.

A Pytorch-based BERT model was fine-tuned on the training dataset with a batch size of 32, a learning rate of 0.00004, a max sequence length of 128, an AdamW optimizer, and the model type was configured as 'Base' (768 hidden units, also the length of each token embedding vector). The best model was obtained after being fine-tuned for 8 rounds, and the precision, recall and F1-score, calculated by Equations (11-13), has reached 78.5%, 79.1% and 78.8% respectively, which indicate that the model is suitable for performing NER on the whole dataset.

$$precision = \frac{TP}{TP + FP} \quad (11)$$

$$recall = \frac{TP}{TP + FN} \quad (12)$$

$$F1 = \frac{2 \times precision \times recall}{precision + recall} \quad (13)$$

where TP represents the number of correctly recognised entity words, FP represents the number of non-entity words that are wrongly recognised as entity words and FN represents the number of unrecognised entity words.

4.2.2. GAT

After applying the BERT model for NER, a sub-dataset consisting of 3623 patents with 8013 citation relationships was derived, with each patent containing both Function and Technology entities and being tagged as 'Function-Technology'. Using the citation network formed by this sub-dataset, we trained a Pytorch-based GAT model on a node classification task, with a learning rate of 0.0018, a dropout rate of 0.1, 4 attention mechanisms and 150 hidden units, and the best model was obtained after 350 rounds of training.

4.2.3. Models in baselines

Since our approach incorporating both text representation learning and graph representation learning, in the baseline methods, we introduced Bag of Words (BoW), Doc2Vec, RoBERTa, ALBERT, graph convolutional network (GCN) and GraphSAGE models for comparative experiments. The BoW model is a traditional text representation approach that encodes each patent based on the presence or absence of individual words using one-hot encoding scheme. Doc2Vec is an unsupervised machine learning algorithm for text representation learning, which can generate patent-level embedding vectors directly. The Doc2Vec model used in our experiments was trained for 12,000 rounds with a vector length of 160, a window size of 15 and a learning rate decaying from 0.001–0.0001. RoBERTa and ALBERT are two variants of BERT, each with certain differences in structure or pre-training tasks compared to BERT, and both of them were fine-tuned using the same hyper parameters and datasets as BERT. GCN and GraphSAGE are two graph embedding models that adopts convolutional operations and sampling & aggregation operations, respectively. In our experiments, both the GCN and GraphSAGE models were trained using the same sub-dataset and hyper parameters as in training the GAT model.

Table 1. Results of the patent classification experiment.

Method	F1-score
BoW-GAT	0.423
Doc2Vec-GAT	0.481
BERT-GraphSAGE	0.541
BERT-GCN	0.691
RoBERTa-GAT	0.691
AlBERT-GAT	0.727
BERT-GAT (ours)	0.735

4.3. Patent classification

The patent classification experiments were conducted to evaluate the representation capabilities of the methods for Function and Technology dimensions after introducing citation relationships. Among the six baseline methods, which were BoW-GAT, Doc2Vec-GAT, RoBERTa-GAT, ALBERT-GAT, BERT-GCN and BERT-GraphSAGE, the first four serve as benchmarks for the representation learning of patent contents, while the other two are used as benchmarks for the representation learning of the citation network. For the evaluation metrics, considering the significant differences in the number of patents corresponding to each feature tag in the sub-dataset, we employed precision, recall and F1-score under micro averaging mode. As precision, recall and F1-score are identical when applying the micro averaging mode due to the counterbalancing judging criteria of *FP* and *FN* in multiclass classification tasks, we simply recorded the F1-scores.

Table 1 shows the results of this experiment. While all the three approaches have the ability of text representation, BERT-related models outperformed BoW and Doc2Vec in terms of semantic exploitation ability, indicating that the finer granularity of text representation and the focus solely on terms enabled our method to filter out noises. Additionally, the BERT model demonstrated superior performance compared to its two variants. When integrating the Citation dimension through the graph embedding process, our method retained the features of both the Function and Technology dimensions more effectively because of the introduction of attention mechanisms, compared with the GCN and GraphSAGE included methods. Therefore, our method shows a notable advantage in representing the Function and Technology knowledge dimensions of patents.

4.4. Citation prediction

In an effort to evaluate our method's ability in integrating the Citation dimension, a citation prediction experiment was carried out, in which we regarded the recommendation list as a predicted citation list and the actual citation list as the ground truth. In this experiment, we introduced BoW-GAT, Doc2Vec-GAT, RoBERTa-GAT, ALBERT-GAT, BERT-GCN and BERT-GraphSAGE as baselines again, and the initial patent vectors without graph representation learning were used as a blank control group. As two commonly used ranking performance indicators, we selected normalised discounted cumulative gain (NDCG) and mean average precision (mAP) and conducted evaluations on predicted citation lists of different lengths, specifically for each patent, if more actual neighbours appeared in its predicted citation list, both of the two indicators would also be higher. These two indicators could be calculated

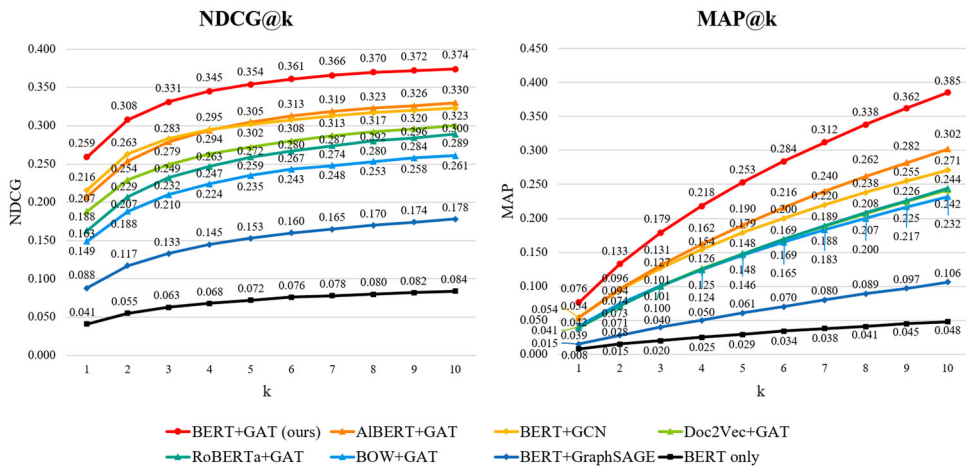


Figure 7. Results of the citation prediction experiment, with k ranging from 1 to 10.

by Equations (14-15), and the results are presented in Figure 7.

$$NDCG = \frac{\sum_{i=1}^k \frac{1}{\log_2(i+1)}}{\sum_{i=1}^{TP} \frac{1}{\log_2(i+1)}} \quad (14)$$

$$AP = \frac{\sum_{i=1}^k \frac{TP@i}{i}}{n_t} \quad (15)$$

where i represents each patent in the predicted citation list, k is the pre-assigned length of the predicted citation list, TP is the number of correctly predicted patents, $TP@i$ refers to the temporary TP value when the summation operation reaches i , n_t is the length of the actual citation list and the mAP value could be calculated by the mean of the average precision (AP) values of all the patents. It is evident that the citation lists predicted by our BERT-GAT method are closer to the actual lists, thus also having a significant advantage in representing the Citation dimension compared with the other two methods. Furthermore, both indicators show a rising trend as the length of the predicted citation lists increases, suggesting that longer prediction lists can reflect the Citation dimension more comprehensively.

5. Application

In this section, two application cases about sealing structure design are demonstrated to validate the feasibility of the proposed methodology from a microscopic and a macroscopic view respectively. The description of the knowledge recommendation process and the analysis of recommendation results illustrate the establishment of the knowledge requirement space and the innovative design ideas generated from recommended patents. The results of a user survey shows the effectiveness of knowledge recommendations with two mechanisms from the designers' perspective. A recommender system prototype was developed for the implementation, and the interface is shown in Figure 8. Interface of the

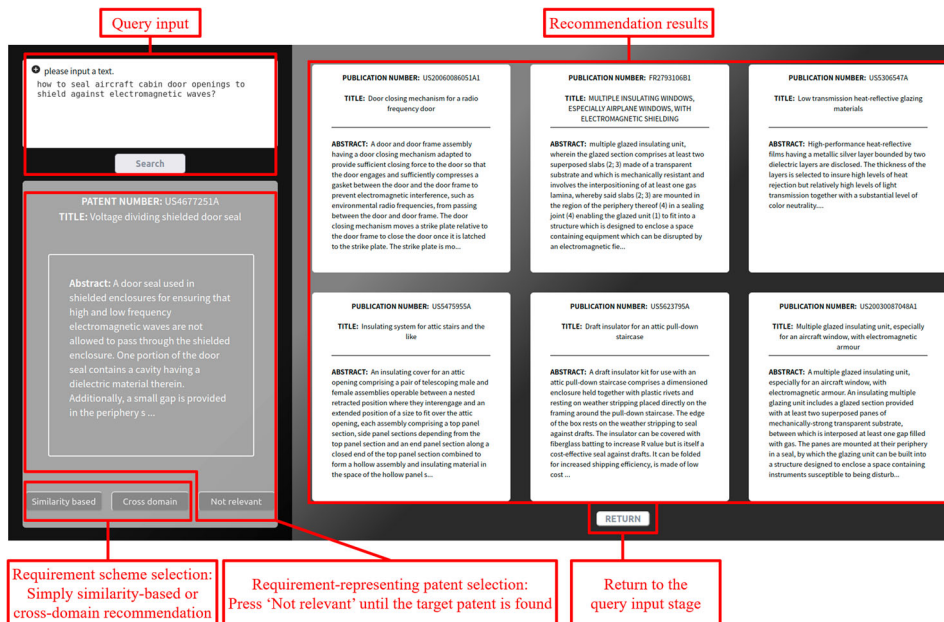


Figure 8. Interface of the prototype recommender system. An example of cross-domain knowledge recommendation is demonstrated.

prototype recommender system. An example of cross-domain knowledge recommendation is demonstrated. It should be noted that the recommendation results presented in this paper only include publication numbers, titles and IPC codes, since the texts of abstracts are lengthy and not suitable for being accommodated in tables.

5.1. Knowledge recommendation process with result analysis

From a microscope perspective of the contents of recommendation results, to elaborate on the recommendation process and how the results inspire innovative design ideas, a knowledge recommendation task about the electromagnetic sealing structure design on aircraft cabin door openings is carried out, and examples of the recommendation results are analyzed. Aircrafts commonly encounter various harmful electromagnetic waves during flight, originating from solar radiation, lightning and various electric devices. As a crucial component connecting the external and internal environments of an aircraft, the electromagnetic sealing performance of the aircraft cabin door is especially vital for flight safety and the proper functioning of communication systems. However, different from ordinary engineering products, aircraft cabin doors are characterised by intricate and delicate structures, making their design process reliant on abundant technical knowledge. Therefore, in the innovative design process of the electromagnetic sealing structure on aircraft cabin door openings, providing designers with patent knowledge, especially cross-domain knowledge, is necessary for the generation of innovative solutions.

5.1.1. Requirement-representing patent selection

According to the design requirements, we entered a search query into the prototype system: ‘How to seal aircraft cabin door openings to shield against electromagnetic waves?’ After performing NER and normalisation process on the query, the system pushed the most relevant patents sequentially, and in this case, we selected the first recommended patent (US4677251A) as a representation of the knowledge requirements. At the same time as the system processing, we also checked the entities recognised in the query, which are ‘cabin-Technology_Opening’, ‘door opening-Technology_Opening’ and ‘electromagnetic waves-Function_Electromagnetic&ThermalProtection’. These entities indicates that the representation learning process correctly captured our knowledge requirements from both the Function and Technology dimensions.

As the requirement-representing patent, Patent US4677251A is titled ‘Voltage dividing shielded door seal’, and its abstract describes an electromagnetic shielding door seal containing a dielectric material-filled cavity and a small gap respectively for absorbing and dissipating electromagnetic waves. Considering the number of codes at different levels in the hierarchical IPC classification system, we adopted the Class-level codes (the first three digits of the full code) in this case, and thus Patent US4677251A has a single IPC code ‘H05’, which represents ‘Electric techniques not otherwise provided for’.

5.1.2. Similarity-based knowledge recommendation

With the selection of the requirement-representing patent, a series of the most relevant patents are also determined. To achieve the best recommendation performance, we set the length of the recommendation list to 6, and the results solely based on cosine similarity are presented in Table 2. It is apparent that most recommended patents contain the IPC code H05, and only two other IPC codes appeared in the results, suggesting that the similarity-based recommendation results are generally within the same domain as the requirement-representing patent.

Taking Patent US4910920A as an example, what described in this patent is an electromagnetic sealing structure formed by alternating arranged conductive resilient finger stocks and absorbing material, which can provide shielding effect against electromagnetic waves of a large range of frequencies. Similar to the requirement-representing patent, this patent also adopts energy absorption as a means to protect against electromagnetic waves, but it introduces alternative options in the implementation.

5.1.3. Cross-domain knowledge recommendation

For cross-domain knowledge recommendation, we assigned the threshold for Jaccard similarities of IPC code lists as 0.5, and the resulting recommended patents are presented in

Table 2. Similarity-based recommendation results.

Publication Number	Title	IPC Code
US4910920A	Electromagnetic door seal	H05
US5214242A	Electromagnetic interference/radio frequency interference conducting strip	H05
EP0269206A1	A door seal for excluding electromagnetic interference	H05, E06
US4817337A	Radiation resistant door seal	H05, E06
US4069618A	Radiant resistant closure assembly	H05
US20060086051A1	Door closing mechanism for a radio frequency door	E05

Table 3. Cross-domain recommendation results.

Publication Number	Title	IPC Code
US20060086051A1	Door closing mechanism for a radio frequency door	E05
FR2793106B1	Multiple insulating windows, especially airplane windows, with electro-magnetic shielding	H05, B64, E06, C03
US5306547A	Low transmission heat-reflective glazing materials	B32, E06, C03
US5475955A	Insulating system for attic stairs and the like	E04
US5623795A	Draft insulator for an attic pull-down staircase	E04
US20030087048A1	Multiple glazed insulating unit, especially for an aircraft window, with electromagnetic armour	H05, B64, E06, C03

Table 3. Among the 6 patents, 7 different IPC codes have appeared, and only 2 of the patents have IPC code lists containing H05, suggesting that the recommendation results have covered a wide range of domains. To demonstrate that recommendation results from a wider range of domains can better assist designers in generating innovative ideas, we chose two patents from the cross-domain recommendation list for illustration.

The first example is Patent FR2793106B1, which is titled ‘Multiple insulating windows, especially airplane windows, with electromagnetic shielding’. This patent disclosed a window glazing insulating structure including multiple gas layers, with conductive laminae and connectors connecting between and around each layer. The innovative idea presented in this patent is redundancy design, specifically employing multiple shielding units instead of a single one, thereby enhancing the stability and reliability of electromagnetic protection. When a shielding unit fails, the remaining units can still provide electromagnetic shielding effect.

The second example is Patent US5306547A, which is titled ‘Low transmission heat-reflective glazing materials’. What introduced in this patent is a heat-reflective film comprising a silver layer and two dielectric layers, and each layer has a specific thickness. Currently, most commonly used electromagnetic shielding structures on aircraft cabin doors are based on absorption or dissipation mechanisms, while this patent provides a novel perspective of reflection. Absorbing or dissipating electromagnetic waves usually requires additional energy consumption, and the materials are susceptible to damage or aging. In contrast, reflection-based methods prevent the energy of electromagnetic waves from entering the interior of the material, resulting in better durability and energy efficiency.

5.2. User survey on engineering design tasks

From a macroscopic perspective of designers, to assess whether the proposed method and the recommender prototype can meet their needs, we conducted a survey on three engineering design tasks with 30 participants. The content of the three design tasks are all relevant to sealing structures, which are outlined in Table 4. All participants recruited for this survey were master or doctoral students majoring in industrial engineering and had no prior experience in sealing structure design, ensuring a relatively consistent level of relevant knowledge. Among the 30 participants, half of them used the similarity-based recommendation scheme while the other half used the cross-domain recommendation scheme.

During the survey, participants were asked to input three queries in the role of designers for each design task. Once the system presented the recommendation results, participants

read the titles and abstracts of the 6 patents and then gave comprehensive ratings from 0 to 10 for whether the top 1, top 3 and top 6 recommendation results provided design ideas. In addition, participants were also suggested to note down their design ideas at the beginning and end of each task. These design ideas were transformed into corresponding entity tag lists through NER, and the Jaccard similarities (Equation 10) between these lists and the one corresponding to the description of the current design task were calculated, as the relevance of ideas to the design task. The change in the relevance before and after knowledge recommendation reflects the impact of knowledge recommendation on the generation of design ideas.

Tables 5 and 6 present the results of user ratings and the relevance of design ideas to the tasks, respectively. Some examples of design ideas after knowledge recommendation are listed in Table 7. In general, the average user ratings are around 6, indicating that the recommendation results can meet designers' needs from their subjective viewpoint. Meanwhile, the ascending trend in the recommendation results for the top 1, top 3 and top 6 implies that more recommendation results would be more likely to provide design ideas. Comparing before and after the knowledge recommendation, the relevance between participants' design ideas and the three design tasks has all shown an increase, with an average growth rate of 21.0%. This result objectively demonstrates that with the assistance of our prototype system, designers can generate more reasonable design ideas.

6. Conclusion

Facing the problem of insufficient patent representation learning and the arbitrary patent knowledge recommendation results, it is still of vital importance to construct an effective

Table 4. Outline of three engineering design tasks in the user survey.

Task	Description
Task 1	Aircraft cabin door sealing structure that ensures airtightness and provides electromagnetic shielding capability.
Task 2	Aircraft fuel tank sealing structure that protects against fire and lightning strikes.
Task 3	Automobile sunroof seal that prevents the penetration of rain water and dust.

Table 5. User ratings for the recommendation results.

	Similarity-based			Cross-domain		
	Top 1	Top 3	Top 6	Top 1	Top 3	Top 6
Design task 1	6.13	6.50	6.54	6.18	6.73	6.69
Design task 2	6.29	6.25	6.08	5.47	6.00	6.49
Design task 3	5.81	6.52	6.98	5.51	6.16	6.78
Average	6.08	6.42	6.53	5.72	6.30	6.65

Table 6. Relevance of design ideas to the design tasks.

	Before recommendation	After recommendation	Growth
Design task 1	0.451	0.549	21.7%
Design task 2	0.573	0.635	10.8%
Design task 3	0.479	0.635	32.6%
Average	0.501	0.606	21.0%

Table 7. Examples of design ideas after knowledge recommendation.

Design task 1	Foam sealing devices can be used to ensure air tightness, and honeycomb structures can be designed for electromagnetic protection.
Design task 2	Resistance and capacitance measuring instrument with electrostatic coating on the surface of anti-spark lightning cabinet; using sulphur polymer, carbon fibre polymer and different apertures to ensure air tightness.
Design task 3	Use sealing strips and glass with defined channels to facilitate drainage, while filling with highly absorbent materials to prevent dust.

patent representation learning approach to structurally extract the key technical information in patent documents. This study proposed a BERT-GAT-based patent representation learning approach to effectively explore knowledge from patent documents, in an effort to support designers in innovative engineering design. The contributions of this study include the following three aspects.

- A four-dimensional patent knowledge model is defined for the formalised representation of patent knowledge across different types, including Domain, Function, Technology and Citation. It presents as a theoretical model for organising extensive and various patent knowledge orderly. With the knowledge model, patent knowledge from different dimensions could be processed separately, and interference between them could be avoided. Meanwhile, the knowledge model could also assist in accurately linking different dimensions of patent knowledge to designers' diverse knowledge requirements, such as cross-domain knowledge, and the effectiveness of knowledge recommendation could be improved.
- A deep learning-based patent knowledge representation learning approach consisting of BERT and GAT is proposed to comprehensively exploit patent knowledge from textual contents and the citation network. By applying a fine-tuned BERT model for NER, knowledge embedded in patent texts could be deeply mined. The utilisation of the GAT model could further integrate knowledge from citation relationships into patent representations, thereby providing designers with recommendations with more abundant knowledge. Different from other patent representation learning approaches, this approach provides a solution to the difficulty of fully extract and integrate patent knowledge from both patent texts and citation relationships, thus improving the depth of knowledge in patent representations.
- A recommender system prototype with different recommendation schemes was developed, which could serve as an auxiliary tool in innovative engineering design. Unlike conventional patent retrieval and recommendation tools, our prototype provides functions of both simply similarity-based and cross-domain knowledge recommendation. The former one can realise the recommendation of patents that most relevant to the knowledge requirements from all aspects, whereas the latter can provide designers with related patents from various domains, mitigating the limitation of sticking to a fixed mindset in the design process. Meanwhile, in the application cases, the recommendation results show that our patent knowledge representation and recommendation method as well as the recommender prototype could effectively facilitate the generation of innovative design ideas. The application cases can serve as a practical example, illustrating the cohesive collaboration of between knowledge representation learning with recommendation and innovative engineering design.

However, some limitations still exist. On the one hand, the proposed method has only been validated in the design of sealing structures, with no application yet in larger-scale engineering design scenarios for patent representation learning and knowledge recommendation. On the other hand, apart from textual content and citation relationships, patents' metadata, e.g. publication date and inventor information, also contain valuable knowledge, and appropriately integrating them in the representation learning process would contribute to a more flexible knowledge recommendation. As a result, our future work will mainly focus on expanding the scale of representation learning and constructing more extensive datasets for training and validation. Meanwhile, due to the outstanding performance of large language models (LLMs) in various text analysis tasks, incorporating LLMs into our methodology to enhance patent text representation is also a direction of our future research.

This study strives to provide an innovative solution that bridges the gap between patent knowledge representation and innovative engineering design, ultimately enhancing the creative process and fostering the generation of ground-breaking design concepts.

Disclosure statement

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